

Zhiheng (Kevin) Li

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EDUCATION

University of Minnesota, Twin Cities — Minneapolis, MN Expected May 2028
Bachelor of Science in Mathematics · Computer Science Minor · GPA: 4.0/4.0
Honors: Dean's List (Fall 2025, Spring 2026) · Tau Sigma Honor Society
Relevant Coursework: Calculus I & II (A), Linear Algebra & Differential Equations (A), Physics for Scientists & Engineers I & II (A), Intro to Programming Concepts — Python (A), Statistical Analysis (A)

RESEARCH EXPERIENCE

SuperCDMS Dark Matter Experiment · School of Physics & Astronomy, UMN Oct 2025 – Present
Undergraduate Research Assistant & UROP Fellow · Advisor: Prof. Yan Liu · Postdoctoral Mentor: Shubham Pandey

- Generated 1×1 and N×M phonon pulse templates for all SuperCDMS SNOLAB R4 detectors (up to 10 Ge detectors, 12 phonon channels each) using PCA-based component extraction and 4-exponential model fitting; delivered ROOT files for downstream optimal filter reprocessing ahead of the June 18 deadline (June 2026).
- Migrated a 2.1 GB PostgreSQL database (12 M+ rows, 13 tables) from the BlueFors CS2 Windows control PC to a 12.7 TB Linux server; designed and deployed BlueFors-Alert, an interactive Slack bot providing real-time sensor threshold alerts and remote query commands without server access (June 2026).
- Collaborated with postdoctoral researcher Shubham Pandey on biweekly progress reviews; integrated analysis methods into the experiment's reconstruction workflow; conducted data analysis and project development on the MSI Agate HPC cluster.
- Built a Python data pipeline in Jupyter Notebook to process LED calibration data from 11 cryogenic Ge detector channels: baseline subtraction, SNR-based glitch detection, peak alignment, and exponential fitting over 5,200+ events (10+ GB raw data); explored ML approaches for glitch classification (Oct 2025).
- Extended RuOx thermometer calibration curves (R31279 & R31839) from 7 mK–102 K to the full 5 mK–300 K operational range using PCHIP interpolation + Mott VRH extrapolation; generated validated Lake Shore .340 files for direct temperature controller loading (May 2026).
- Participated in the physical installation of a ~\$500K BlueFors XLD dilution refrigerator (May 2026); set up compressor water-cooling, assembled a 4K heater for diagnostics, and identified elevated P5 valve pressure as the root cause of temperature anomaly.

SELECTED PROJECTS

Novel Billiards Rest (Bridge Rest) · Entrepreneurship Project — Beijing, China Oct 2024 – Jun 2025
Team Leader

- Led a student team to design a cost-efficient billiards rest prototype resolving a grip redundancy problem in professional play; received national-level project approval and seed funding; Chinese Utility Patent application (No. 202520618681.1) submitted to CNIPA — currently under review.
- Invited to present at Tsinghua X-Lab & Kering Entrepreneurship Roadshow; presented to Ms. Cai Jinqing, President of Kering Greater China, receiving high-profile industry endorsement.

Low-Carbon Transition Project · China Steel Industry — Beijing, China Oct 2024 – May 2025
Undergraduate Researcher

- Assisted in research on low-carbon transition pathways for China's steel industry; contributed to data collection, database construction, and quantitative analysis for national low-carbon initiatives.

AWARDS & HONORS

UROP Fellowship, University of Minnesota	Summer 2026
First Prize, Beijing University Student Innovation & Entrepreneurship Competition (Science & Technology Track)	2025
First Prize, National College E-Commerce "Innovation, Creativity & Entrepreneurship" Challenge — Team Leader	2025
Third Prize, China International College Students' Innovation Competition — Team Leader	2025
Second Prize, NECCS National English Competition for College Students	2024

ACTIVITIES & INVOLVEMENT

Chinese Student Union, UMN — Active Member	Oct 2025 – Present
CSSA, UMN — Volunteer Calculus I Tutor	2025 – Present
College Table Tennis / Football / Guitar Club — Team Member	2023 – 2024

PROFESSIONAL SKILLS

Techniques:	Cryogenics · Signal Processing · Data Pipelines · Thermometer Calibration · Database Migration · Slack Bots
Tools:	Python · PostgreSQL · NumPy/Pandas · SciPy · Matplotlib · Git/GitHub · Jupyter · MSI HPC (Agate) · ROOT · Java · Excel
Languages:	Mandarin Chinese (Native) · English (Fluent; IELTS 7.0)